Bernoulli and Binomial Distributions — Problem Set

Warm-up

1. A fair six-sided die is rolled. Success is defined as rolling a 6. State the probability of success p and the probability of failure q.

2. State the three conditions required for a random variable to follow a Bernoulli distribution.

3. If $X \sim \text{Bernoulli}(0.4)$, find the expected value $E(X)$ and the variance $\text{Var}(X)$.

4. A trial has a probability of success $p = 0.75$. If 10 independent trials are performed, state the parameters n and p for the resulting binomial distribution.

5. Let $X \sim \text{Bin}(15, 0.2)$. Write the expression for $P(X = 3)$ using the binomial probability formula.

6. A student guesses on a true/false test with 10 questions. What is the probability that the student gets no questions correct?

7. Calculate the mean of a binomial distribution where $n = 50$ and $p = 0.12$.

8. If the variance of a binomial distribution is 9 and $n = 100$, find the value of p (assume $p < 0.5$).

9. For $X \sim \text{Bin}(n, p)$, write the formula for $P(X \geq 1)$ in terms of $P(X = 0)$.

10. Identify which of the following is NOT a binomial experiment:

    a) Counting the number of heads in 20 coin tosses.

    b) Drawing 5 cards from a deck without replacement and counting the number of Aces.

Skill-building

1. A marksman has a probability of 0.8 of hitting a target. If he fires 6 shots, calculate the probability he hits the target exactly 4 times.

2. Let $X \sim \text{Bin}(8, 0.3)$. Calculate $P(X < 2)$ to four decimal places.

3. In a large batch of lightbulbs, 5% are defective. If a sample of 20 bulbs is chosen, find the expected number of defective bulbs and the standard deviation.

4. If $X \sim \text{Bin}(n, p)$ has a mean of 10 and a variance of 8, determine the values of n and p.

5. The probability of a computer chip being faulty is 0.02. Find the probability that in a box of 50 chips, at least one is faulty. [cite: 8, 12]

6. A binomial distribution has $n = 4$ and $P(X = 4) = 0.0625$. Find the value of p and hence find $P(X = 2)$.

7. Given $X \sim \text{Bin}(100, 0.5)$, use the properties of the distribution to find $E(X)$ and $\text{Var}(X)$.

8. A fair coin is tossed 10 times. Compare the probability of obtaining exactly 5 heads with the probability of obtaining exactly 4 heads.

9. For a binomial random variable X, it is known that $E(X) = 12$ and $\mathrm{Var}(X) = 3$. Is it possible to find n? If so, calculate it.

10. Suppose $X \sim \mathrm{Bin}(n, p)$. If $n = 2$ and $P(X = 0) = 0.49$, find $P(X = 1)$.

Easier Exam Questions

1. (Manly 2020 Q2) Which one of the following describes a Bernoulli random variable?

   (A) The times achieved by athletes in a 100-metre race
   (B) The value of a card chosen from a pack of 52 playing cards
   (C) Choosing a red pen from a container that contains red, black and blue pens
   (D) The number of emails a doctor receives in a week

2. (Moriah 2021 Q1) An examination consists of 20 questions, each having four possible answers. A student guesses the answer to every question. Let X be the number of correct answers. What is the value of $E(X)$?

   (A) 4
   (B) 5
   (C) 10
   (D) 16

3. (Blacktown Boys 2022 Q3) A coin is biased such that the probability of a head on any toss is 0.69. Which expression states the probability of a head appearing twice on this coin if this coin is tossed 50 times?

   (A) $\binom{50}{2} \times 0.31^2 \times 0.69^{48}$
   (B) $50 \times 0.31^2 \times 0.69^{48}$
   (C) $\binom{50}{2} \times 0.69^2 \times 0.31^{48}$
   (D) $50 \times 0.69^2 \times 0.31^{48}$

4. (Glenwood 2022 Q8) The probability that it will rain on any given day in February 2024 is 0.2. What is the probability that February 2024 (a leap year) will have exactly 8 rainy days?

   (A) $1.5 \times 10^{-7}\%$
   (B) 9%
   (C) 10%
   (D) 29%

5. (Glenwood 2020 Q11d) Weather records suggest that in the town of Mathsville, an average of 5 days out of of 30 in the month of April will be wet. **3**

   If X represents the number of wet days in April, such that $X \sim \mathrm{Bin}\left(30, \frac{1}{6}\right)$, find $P(X \geq 2)$.

Harder Exam Questions

1. (Hunters Hill 2022 Q9) In a lollypop factory the material for making each lollypop is sent through a machine. The time, X seconds, taken to produce a lollypop by the machine is a binomial distribution where it can be shown that $P(X \leq 3) = \frac{5}{29}$. A random sample of 12 lollypops is chosen. The probability that exactly 5 of these 12 lollypops took 3 or less seconds to produce, is:

   (A) $\binom{12}{3}\left(\frac{5}{29}\right)^3\left(\frac{24}{29}\right)^{12}$

   (B) $\binom{12}{3}\left(\frac{5}{29}\right)^3\left(\frac{24}{29}\right)^{9}$

   (C) $\binom{12}{5}\left(\frac{5}{29}\right)^5\left(\frac{24}{29}\right)^{12}$

   (D) $\binom{12}{5}\left(\frac{5}{29}\right)^5\left(\frac{24}{29}\right)^{7}$

2. (Glenwood 2021 Q3) The probability of Hannah scoring a goal in a netball game is 0.45. What is the least number of games that must be played to ensure the probability of scoring a goal at least once is more than 0.9?

   (A) 2

   (B) 3

   (C) 4

   (D) 5

3. (Baulkham Hills 2020 Q12a) The following table is the distribution of a random variable X where $X \sim \text{Bin}(4, p)$.

x	0	1	2	3	4
$P(X = x)$	0.2401	a	b	0.0756	0.0081

   (i) Use the known entries in the probability distribution table to calculate the value of p. **1**

   (ii) Find the values of a and b **2**

   (iii) Calculate $\text{Var}(X)$. **1**

   (iv) Find $P(X \geq 3)$. **1**

4. (Blacktown Boys 2023 Q5) The random variable X is such that $X \sim Bin(n, p)$. The mean value of X is 225 and variance of X is 144. What is the value of p?

   (A) 0.8

   (B) 0.64

   (C) 0.6

   (D) 0.36

5. (Knox 2021 Q12b) Weather observations in the town of Dampville have established that the probability of rain on any given day is 0.8.

   Observations are made for 100 consecutive days.

   Let X be the random variable representing the number of rainy days.

(i) Find the expected value $E(X)$. **1**

(ii) Find the standard deviation of X. **1**

(iii) By considering a normal distribution find the approximate probability that **1**

$$76 \leq X \leq 84$$

6. (Caringbah 2020 Q11e) The discrete random variable X has a binomial distribution with n independent trials with probability of success p. If its mean is 6 and its standard deviation is 2, find the values of n and p. **3**

7. (Newcastle Grammar 2020 Q13d) A multiple-choice test contains ten questions. Each question has four choices for the correct answer. Only one of the choices is correct.
(i) What is the probability of getting 70% correct with random guessing? **1**
(ii) What is the probability of getting at most 70% correct with random guessing? **2**